

Question

A quantitative analyst is building an NLP pipeline to extract sentiment from quarterly earnings call transcripts. She intends to apply five preprocessing steps in sequence: (1) lemmatization, (2) tokenization, (3) stop-word removal, (4) feature extraction, (5) lowercasing. To obtain a valid and standard pipeline, in what order should these steps be applied?

- A. Any order works; preprocessing steps are independent.
- B. Tokenization → lowercasing → stop-word removal → lemmatization → feature extraction.
- C. Feature extraction → tokenization → lowercasing → lemmatization → stop-word removal.
- D. Lemmatization → stop-word removal → tokenization → feature extraction → lowercasing.

Answer: B

Explanation

The standard pipeline is: tokenize (split into words), lowercase (normalize case), remove stop words (drop low-info tokens), lemmatize (reduce to base forms), and finally extract numeric features. Option A is wrong because order matters. Option C reverses the sequence. Option D applies lemmatization before tokenization, which is impossible without tokens.

Question

A risk team builds a sentiment classifier using a binary bag-of-words representation of news articles about a firm. The model performs well on formal news but struggles with sarcastic social media posts. Which of the following best explains this limitation?

- A. The vocabulary is too small.
- B. Bag-of-words ignores word order and context, so sarcasm, negation, and contextual reversal of meaning are lost in the vector representation.
- C. The model requires more data.
- D. The binary representation ignores word frequencies.

Answer: B

Explanation

BoW treats each word independently and discards order and context. Sarcasm depends on context and word co-occurrence patterns that BoW cannot capture. Options A, C, and D are less fundamental — even with large vocabularies and count-based BoW, the context problem persists.

Question

A risk analyst computes TF-IDF scores across 1,000 earnings call transcripts. The word “million” appears in almost every document; the word “impairment” appears in a small subset. What will TF-IDF assign to each word, and why?

- A. High score to both words because both are common.
- B. High score to “impairment” (high IDF due to rarity), low score to “million” (low IDF because it appears in most documents).
- C. High score to “million” (high frequency), low score to “impairment”.
- D. Equal scores because TF-IDF normalizes frequencies.

Answer: B

Explanation

TF-IDF increases weight for terms that are frequent in a document but rare across the corpus. “Million” is ubiquitous and therefore uninformative (low IDF); “impairment” is rare and thus discriminative (high IDF). Options A, C, and D misinterpret TF-IDF.

Question

A sentiment classifier misclassifies the phrase “not bad” as negative because it identifies “bad”. Which NLP technique would most directly address this issue?

- A. Lemmatization.
- B. Stop-word removal.
- C. Bi-grams (or higher-order n-grams) that treat “not bad” as a single unit.
- D. Dimensionality reduction.

Answer: C

Explanation

N-grams preserve local word order, allowing the model to treat phrases like “not bad” or “no loss” as single features. This addresses negation explicitly. Options A and B do not capture context. Option D is unrelated.

Question

Using pre-trained word embeddings, an analyst computes cosine similarity between the following pairs: (a) “bank” and “financial”; (b) “bank” and “tree”; (c) “bank” and “river”. Which is likely to have the highest cosine similarity and why?

- A. “bank”–“tree” because both are nouns.
- B. “bank”–“financial” if the embedding was trained on financial text, because “bank” in that context appears in similar semantic neighborhoods as “financial”; embeddings capture context-dependent meaning.
- C. “bank”–“river” because both are physical locations.
- D. All three pairs will have equal similarity.

Answer: B

Explanation

Word embeddings are trained on large corpora and capture distributional context. In a financial corpus, “bank” co-occurs with “financial” and thus has high similarity. The polysemy of “bank” (financial institution vs river bank) is captured by distinct embedding neighborhoods depending on the training corpus. Options A, C, and D ignore context.

Question

In the sentence "The trader liquidated the position because it breached the risk limit", a transformer model correctly associates "it" with "the position" rather than "the trader". Which mechanism enables this?

- A. Random initialization.
- B. Self-attention, which computes pairwise relevance scores between all tokens and allows the model to weight the contribution of "the position" more heavily when processing "it".
- C. Stop-word removal.
- D. Lemmatization.

Answer: B

Explanation

Self-attention learns to weight each token's contribution to every other token based on relevance. This enables coreference resolution and long-range dependency modelling. Options A, C, and D do not explain the mechanism.

Question

A legal analyst uses an LLM to review a 500-page contract. The LLM has a context length of 32,000 tokens. The analyst pastes the entire contract but the model's responses are truncated and unreliable. What is the most likely issue?

- A. The model's temperature is too high.
- B. The document exceeds the effective context length; even if the maximum is nominally 32,000 tokens, effective performance degrades for very long inputs. The analyst should use chunking, retrieval-augmented generation (RAG), or a model with larger context.
- C. The contract has too much text.
- D. The model cannot process PDFs.

Answer: B

Explanation

Context length limits are the maximum tokens the model can process in a single input/output sequence. Even large context windows lose effectiveness at extreme lengths. Chunking or RAG are the standard remedies. Options A, C, and D are secondary or irrelevant.

Question

A risk analyst uses an LLM to generate written commentary on VaR breaches. Setting temperature high yields creative but inconsistent outputs; setting temperature low yields repetitive but predictable outputs. The analyst wants factual, consistent outputs. What is the most appropriate setting?

- A. High temperature.
- B. Low temperature (e.g., 0.1–0.3) to narrow the sampling distribution, and consider using a moderate Top-P to exclude implausible tokens while retaining some diversity.
- C. Set temperature to 2.0.
- D. Ignore these parameters; they have no effect.

Answer: B

Explanation

Low temperature sharpens the probability distribution, favouring high-probability tokens and reducing randomness. Top-P (nucleus sampling) complements this by excluding the long tail. Option A increases randomness. Option C is extreme. Option D is wrong — these parameters strongly influence output.

Question

A compliance officer asks an LLM to cite regulatory guidance on LIBOR transition. The model confidently cites a specific paragraph of a Federal Reserve publication, but on checking, the paragraph does not exist. What is this phenomenon called, and why does it occur?

- A. Data drift.
- B. Hallucination — the model generates plausible but factually incorrect content because it optimises for fluency and statistical plausibility rather than factual accuracy.
- C. Overfitting.
- D. Sampling error.

Answer: B

Explanation

Hallucinations are a well-documented LLM failure mode: outputs are syntactically and stylistically plausible but factually wrong. This occurs because LLMs are trained to predict probable next tokens, not to verify facts. Options A, C, and D are distinct phenomena.

Question

An insurance company wants an LLM to answer questions about its internal claims-handling manual. The manual is proprietary and not in any public training data. Which approach is most appropriate?

- A. Fine-tune the LLM on the manual.
- B. Use retrieval-augmented generation (RAG): index the manual, retrieve relevant passages at query time, and pass them to the LLM as context.
- C. Use a larger LLM.
- D. Increase temperature.

Answer: B

Explanation

RAG is specifically designed for this scenario: proprietary, up-to-date, or specialised knowledge is retrieved from a knowledge base and provided as context to the LLM. Fine-tuning is expensive and can become stale; a larger LLM does not solve the missing data issue; temperature is unrelated.

Question

A risk team needs to extract structured risk categories from unstructured news articles. An analyst uses an LLM with a simple prompt: "What risk categories are in this article?" The outputs are inconsistent. Which prompt engineering improvement is most likely to help?

- A. Increase temperature.
- B. Provide a clear role, specify the desired output format (e.g., JSON), and give a few-shot example showing the expected structure.
- C. Reduce context length.
- D. Use a different language.

Answer: B

Explanation

Effective prompt engineering includes role specification, output format, and few-shot examples. These reduce ambiguity and variance in outputs. Option A increases randomness. Options C and D are unlikely to address the underlying ambiguity.

Question

An analyst asks an LLM to solve a break-even analysis problem. With a direct question, the model gives the wrong answer. With the instruction “Let’s think step by step”, the model arrives at the correct answer. Why does this help?

- A. The instruction is a magic phrase.
- B. Chain-of-thought prompting encourages the model to generate intermediate reasoning steps, which improves multi-step arithmetic and logical tasks by giving the model more tokens to compute intermediate quantities.
- C. The model has more time to compute.
- D. The instruction changes the model weights.

Answer: B

Explanation

Chain-of-thought prompting uses the model’s own token generation to perform intermediate reasoning. Multi-step problems benefit from this because the model can attend to its own intermediate steps. Option A trivialises the mechanism. Option C is inaccurate. Option D is wrong (no weight updates occur).

Question

An AI system autonomously schedules meetings, sends invitations, and reschedules when conflicts arise, using an LLM to reason about calendar constraints. What best describes this system?

- A. A statistical model.
- B. An agentic AI system that uses an LLM to take actions in an environment with limited supervision.
- C. A rule-based classical AI system.
- D. A reinforcement-learning environment.

Answer: B

Explanation

Agentic AI extends LLM capabilities by allowing the model to take actions (send invitations, reschedule) in an environment. Options A, C, and D misclassify the system.

Question

A user has a long conversation with an LLM. After clearing the session and starting fresh, the model has no memory of the previous conversation. What property of LLMs does this illustrate?

- A. Overfitting.
- B. Statelessness — LLMs do not retain information between prompts unless the previous context is explicitly re-supplied.
- C. Regularization.
- D. Data leakage.

Answer: B

Explanation

LLMs are stateless by design: each prompt is processed independently. Maintaining conversation history requires re-supplying prior turns. Options A, C, and D are unrelated.

Question

A vendor claims their LLM achieves 95% on MMLU. Which of the following is a valid caveat about this claim?

- A. MMLU is a comprehensive benchmark that covers all possible tasks.
- B. MMLU scores can be inflated by data leakage (if the benchmark dataset was included in the training corpus), and benchmark saturation means top models cluster near the maximum, reducing discrimination.
- C. MMLU cannot be used to compare models.
- D. MMLU is a qualitative metric.

Answer: B

Explanation

Benchmark caveats include data leakage and saturation. Benchmarks are useful but not sufficient. Options A, C, and D are inaccurate.

Question

A bank deploys an LLM-based chatbot to answer customer inquiries. After several months, the bot starts giving subtly incorrect answers on loan eligibility rules, though it remains fluent and confident. What is the most likely cause, and what should be done?

- A. Random failure; restart the model.
- B. Drift or contamination of the retrieval corpus — if the bot uses RAG, the underlying knowledge base may have become outdated, or the model may have been updated by the vendor. Regular validation and monitoring of GenAI outputs are essential.
- C. The temperature is too low.
- D. The prompt is wrong.

Answer: B

Explanation

GenAI systems require continuous monitoring because outputs are non-deterministic and dependent on retrieval data and vendor updates. Drift in the retrieval corpus or model version can cause subtle errors. Options A, C, and D are less likely given the scenario.

Question

In a model risk hierarchy, which of the following should be classified as a *higher-risk* model based on the standard materiality-complexity-exposure framework?

- A. A simple Excel spreadsheet used by an intern for ad-hoc calculations.
- B. A complex neural network that determines regulatory capital requirements for the bank's trading book.
- C. A single-variable linear regression used to forecast office supplies.
- D. A static lookup table used to convert currency codes.

Answer: B

Explanation

Model risk ranking considers materiality (business impact), complexity (methodology), and exposure (usage). A neural network that drives regulatory capital is high in all three dimensions. Options A, C, and D are low-impact, simple, or limited in scope.

Question

A model with no regularization fits the training data perfectly but generalizes poorly. Adding an L1 penalty improves test performance but zeroes out half the coefficients. How would you characterise this trade-off?

- A. Regularization reduces bias and increases variance.
- B. Regularization reduces variance (overfitting) at the cost of increased bias; L1 also produces sparsity, improving interpretability at the cost of potentially dropping weak but real signals.
- C. Regularization has no effect on bias or variance.
- D. Regularization always improves both bias and variance.

Answer: B

Explanation

Regularization trades off bias for variance: it reduces overfitting by shrinking coefficients, at the cost of introducing bias. L1 additionally performs feature selection, improving interpretability. Options A, C, and D misstate the trade-off.

Question

A regulator asks a bank to explain how its credit-scoring model arrives at a specific decision. The model is a gradient-boosted ensemble that outperforms a logistic regression by 3 percentage points AUC. The bank considers swapping in the logistic regression for compliance reasons. Which statement best captures the trade-off?

- A. The bank should always prioritise interpretability.
- B. The bank should always prioritise accuracy.
- C. The bank should weigh accuracy loss against interpretability gain. A common approach is to keep the high-performing model for production and use post-hoc explainability techniques (SHAP, LIME) to satisfy regulators, rather than downgrade to a simpler model.
- D. The bank should use both models simultaneously and choose randomly.

Answer: C**Explanation**

Interpretability and accuracy are often in tension. Modern practice uses high-performing models with post-hoc explainability rather than sacrificing performance. Options A and B are too categorical. Option D is nonsensical.

Question

An analyst trains 100 decision trees on bootstrap samples of the training data and averages their predictions. Each tree is trained on all available features. What is this technique called, and how does it improve performance over a single tree?

- A. Random forest; it introduces feature randomness.
- B. Bagging (bootstrap aggregation); it reduces variance by averaging decorrelated trees that each see different samples of the training data.
- C. Boosting; it corrects sequential errors.
- D. Gradient descent; it optimises weights.

Answer: B

Explanation

Bagging reduces variance by training on bootstrap samples and averaging. Random forest adds feature subsampling. Boosting is sequential. Gradient descent is an optimisation method. Only B correctly identifies the technique described.

Question

A diagnostic algorithm screens for a rare but fatal disease that affects 0.1% of the population. The disease is curable if caught early. In testing the algorithm, which metric should be prioritised, and why?

- A. Specificity, to avoid false alarms.
- B. Sensitivity, because false negatives (missed cases) have fatal consequences, while false positives only lead to additional testing.
- C. Precision, because positives should be reliable.
- D. Accuracy, because it balances all errors.

Answer: B

Explanation

With rare fatal diseases, the cost of a false negative (missing a real case) is catastrophic. Sensitivity = $TP / (TP + FN)$ measures how many actual positives are caught. Option A optimises for rejecting negatives, which is not the priority. Option C emphasises alert reliability. Option D averages over both, hiding the asymmetry in costs.

Question

A fraud-detection model is trained on 10,000 transactions, of which 20 are fraudulent. After deployment, the model performs well on non-fraudulent transactions but misses most fraud. What is the most likely cause?

- A. Measurement bias.
- B. Representation bias.
- C. Data composition bias due to the extreme class imbalance.
- D. Historical bias.

Answer: C

Explanation

Extreme class imbalance (0.2% positives) causes the model to optimise for the majority class. This is data composition bias. Options A, B, and D describe different mechanisms that are not the primary cause in this scenario.

Question

A logistic regression for hiring decisions includes “years of experience”, “certifications”, and “education level” (coded 1–5). The company realises that including education level may unfairly disadvantage experienced candidates with lower education. Which intervention best ensures equal treatment without removing the variable?

- A. Remove the education variable entirely.
- B. Replace education level with its mean at prediction time, ensuring that the coefficient does not influence individual predictions.
- C. Add interaction terms between education and experience.
- D. Convert education to dummy variables.

Answer: B

Explanation

By substituting the mean education level at inference time, the effect of education on predictions is neutralised across all candidates — they are treated as if they had the same education level. Option A removes the variable but leaves its influence partially embedded in correlated features. Options C and D do not achieve the fairness objective.

Question

A model predicts loan approvals. Group A has 30% approval rate; Group B has 30%. Which fairness metric is satisfied?

- A. Equal opportunity.
- B. Predictive rate parity.
- C. Demographic parity.
- D. Individual fairness.

Answer: C

Explanation

Demographic parity requires equal positive prediction rates across groups. Both groups have 30%, so parity is satisfied. Equal opportunity concerns TPR, predictive rate parity concerns precision, and individual fairness concerns similar treatment of similar individuals. Options A, B, and D refer to different criteria.

Question

A regulator demands simultaneous demographic parity, predictive rate parity, and equal opportunity across two demographic groups with different base rates of default. A mathematical result shows these cannot all be satisfied simultaneously when base rates differ. What is the best practical response?

- A. Demand all three and reject any model that does not achieve them.
- B. Choose the fairness metric most aligned with the business context, document the choice, and accept the trade-offs; involve legal and ethics stakeholders in the decision.
- C. Ignore the impossibility result.
- D. Only use demographic parity in all cases.

Answer: B

Explanation

The Kleinberg-Chouldechova impossibility theorem shows that when base rates differ, demographic parity, predictive rate parity, and equal opportunity cannot all hold. Practical response is to select the metric appropriate to the use case, document the trade-off, and engage governance. Options A, C, and D are impractical or incorrect.

Question

A firm uses a third-party deep learning model for credit decisions. The vendor refuses to share the model weights or architecture. The firm must validate the model. Which approach is most appropriate?

- A. Refuse to deploy the model because full transparency is impossible.
- B. Perform black-box validation: test the model extensively on input-output pairs, benchmark against internal challenger models, examine outcome distributions, and assess fairness metrics across subgroups.
- C. Trust the vendor's claims and deploy without validation.
- D. Use the model without validation but document the risk.

Answer: B

Explanation

Black-box validation is standard practice for vendor models. It uses input-output testing, benchmarking, and outcomes analysis. Option A is too restrictive and impractical. Option C skips essential governance. Option D is non-compliant.

Question

A model is described as “interpretable but not explainable”. Which type of model best fits this description?

- A. A deep neural network with SHAP explanations.
- B. A simple linear regression whose coefficients are known, but whose behaviour on specific complex cases is difficult to describe in narrative form.
- C. A black-box ensemble with no diagnostic tool.
- D. A random assignment process.

Answer: B

Explanation

Interpretable = internal logic is understandable (linear regression, decision trees). Explainable = specific decisions can be justified in narrative. A linear regression is interpretable but narratives around specific outlier cases may be complex. Option A is the opposite (explainable, not interpretable). Options C and D do not fit.

Question

A trader uses an AI-driven pricing tool that occasionally produces incorrect quotes. Despite known flaws, the trader rarely overrides the tool's outputs. Which phenomenon is at play?

- A. Confirmation bias.
- B. Automation bias — over-reliance on automated systems, reducing vigilance and independent verification.
- C. Survivorship bias.
- D. Selection bias.

Answer: B

Explanation

Automation bias is the tendency to trust automated systems too readily, leading to reduced verification and occasional costly errors. Options A, C, and D refer to other cognitive or statistical biases.

Question

Which statement best characterises the difference between group fairness and individual fairness?

- A. Group fairness examines statistical properties across demographic groups; individual fairness requires treating similar individuals similarly.
- B. Group fairness concerns one individual; individual fairness concerns populations.
- C. They are the same concept.
- D. Individual fairness is impossible to define.

Answer: A

Explanation

Group fairness examines outcomes across groups (e.g., demographic parity, equal opportunity). Individual fairness, from Aristotle's "treat like cases alike", requires similar individuals to be treated similarly. Options B, C, and D misstate the concepts.

Question

A model excludes race from its input features. Nevertheless, it discriminates along racial lines because a highly predictive feature (e.g., ZIP code) correlates strongly with race. What is this phenomenon called?

- A. Measurement bias.
- B. Proxy discrimination.
- C. Sampling bias.
- D. Historical bias.

Answer: B

Explanation

Proxy discrimination occurs when a feature correlated with a protected attribute carries discriminatory effects even after the protected attribute is removed. Options A, C, and D are distinct sources of bias.

Question

An AI hiring model trained on 20 years of hiring data favours male candidates for senior roles. The underlying data reflects a period when women were structurally excluded from these roles. Which type of bias does this represent?

- A. Sampling bias.
- B. Historical bias — the historical discrimination is encoded in the training data and reinforced by the model.
- C. Measurement bias.
- D. Deployment bias.

Answer: B

Explanation

Historical bias arises when training data reflects past discrimination. Removing features or reweighting samples are common mitigations. Options A, C, and D describe different phenomena.

Question

An AI system ranks candidates and invites the top-ranked for interviews. Over time, the system's training data increasingly reflects past system decisions. Which risk does this represent?

- A. A self-reinforcing feedback loop, where the model's past biases are amplified in successive training cycles.
- B. Random noise.
- C. Measurement error.
- D. Model overfitting to training data only.

Answer: A

Explanation

Feedback loops occur when model outputs influence future training data, reinforcing initial biases. This is a form of compounding historical bias. Options B, C, and D do not capture the dynamic.

Question

Which of the following best distinguishes “interpretability” from “explainability”?

- A. They are interchangeable terms.
- B. Interpretability is about understanding internal model mechanisms; explainability is about providing human-understandable justifications for specific decisions.
- C. Explainability applies only to linear models.
- D. Interpretability is a legal requirement; explainability is optional.

Answer: B

Explanation

Interpretability = internal understanding; explainability = specific-decision justification. Option A conflates them. Options C and D are incorrect.

Question

A bank deploys a chatbot that incorrectly advises customers about fees, causing public complaints. Which is the primary source of reputational risk?

- A. Model accuracy alone.
- B. Privacy breaches.
- C. Customer perception of incompetence or deception combined with regulatory exposure.
- D. Cost overruns.

Answer: C

Explanation

Reputational risk arises from customer perception (incompetence, deception), amplified by regulatory and legal exposure. Option A is only part of the picture. Options B and D describe distinct sources.

Question

Which of the following best captures the societal risk of generative AI?

- A. High compute costs.
- B. Synthetic content at scale facilitating misinformation, deepfakes, and erosion of public trust in information.
- C. Slow training times.
- D. Privacy risk only.

Answer: B

Explanation

GenAI's societal risks include misinformation at scale, deepfakes, and erosion of trust. Option A is a technical cost concern. Options C and D are partial or incorrect.

Question

An AI mental health chatbot recommends a course of action that conflicts with a patient's stated preference. Which ethical principle is most directly violated?

- A. Nonmaleficence.
- B. Beneficence.
- C. Autonomy — the patient's right to make their own decisions is overridden.
- D. Explainability.

Answer: C

Explanation

Autonomy is the principle of respecting individual decision-making. Overriding patient preference violates autonomy. Options A, B, and D are distinct principles.

Question

Model A has equal true positive rates but unequal false positive rates across groups. Model B has equal false positive rates but unequal true positive rates. Which fairness metric does each satisfy?

- A. Both satisfy demographic parity.
- B. Model A satisfies equal opportunity; Model B satisfies a form of predictive parity (though not the standard definition). Neither satisfies demographic parity.
- C. Both satisfy predictive rate parity.
- D. Both violate all fairness measures.

Answer: B

Explanation

Equal opportunity = equal TPR across groups (Model A). Equal FPR across groups is a form of equalised odds but not standard predictive rate parity. Neither satisfies demographic parity (equal positive prediction rates). Option B is the most accurate.

Question

Which statement correctly distinguishes model risk from algorithmic bias?

- A. Model risk refers to errors and misuse that could cause adverse outcomes; algorithmic bias refers to systematic and unfair discrimination against protected groups.
- B. They are synonymous.
- C. Model risk applies only to AI models; algorithmic bias applies only to statistical models.
- D. Model risk is a legal term; algorithmic bias is a technical term.

Answer: A

Explanation

Model risk encompasses all adverse outcomes from model errors or misuse. Algorithmic bias is a subset focused on unfair discrimination. Options B, C, and D are incorrect.

Question

A criminal uses an AI-generated voice clone of a CEO to authorise a wire transfer. Which category of risk does this represent?

- A. Only individual risk.
- B. Only societal risk.
- C. Both organizational/individual risk (fraudulent loss) and societal risk (undermining trust in voice communications).
- D. Measurement bias.

Answer: C

Explanation

Deepfake-enabled fraud harms the targeted organisation and individual victims, and it erodes trust in the integrity of voice communications more broadly. Options A, B, and D are partial or unrelated.

Question

Which of the following is NOT a recognised source of algorithmic bias?

- A. Problem specification errors.
- B. Data collection and composition.
- C. Model development choices.
- D. Increased computational power.

Explanation

The four recognised sources are problem specification, data, model development, and deployment. Computational power is not a source of bias. Options A, B, and C are all real sources.

Answer: D

Question

A data scientist wants to understand which features drive a complex model's prediction on individual cases. Which technique is most appropriate?

- A. Feature importance scores (global).
- B. Shapley values — they distribute the model's prediction among input features based on marginal contributions.
- C. Grid search.
- D. Feature scaling.

Answer: B

Explanation

Shapley values from cooperative game theory provide per-prediction feature attribution. Feature importance is global. Options C and D are unrelated.

Question

Applicants demand to know why their loan applications were rejected. Which response is most appropriate?

- A. Publish the full model code.
- B. Provide meaningful reasons for each rejection, using explainability tools and consistent criteria.
- C. Refuse to explain, citing trade secrets.
- D. Explain only to rejected applicants from certain demographics.

Answer: B**Explanation**

Regulators and customers expect meaningful explanations. Publish the top contributing factors or counterfactual reasoning. Options A, C, and D are impractical, non-compliant, or discriminatory.

Question

Which is the primary benefit of an organisational AI ethics framework?

- A. It guarantees model accuracy.
- B. It reduces regulatory non-compliance risk by establishing clear principles, review processes, and accountability structures for AI development and use.
- C. It eliminates all bias.
- D. It speeds up model training.

Answer: B**Explanation**

An ethics framework provides principles and processes that align AI with regulatory and societal expectations, reducing compliance risk. It cannot guarantee accuracy or eliminate bias. Training speed is unrelated.

Question

Which statement correctly describes a criticism of utilitarianism when applied to AI ethics?

- A. It ignores consequences.
- B. It can justify violating individual rights when doing so produces greater aggregate benefit.
- C. It emphasises character over actions.
- D. It relies on rigid rules.

Answer: B

Explanation

Utilitarianism maximises aggregate welfare and can justify rights violations for greater total benefit. Option A is the opposite. Options C and D describe virtue ethics and deontology, respectively.

Question

Which best describes a deontological ethics framework?

- A. Judges actions by their consequences.
- B. Judges actions by adherence to duties and rules regardless of consequences.
- C. Emphasises virtuous character traits.
- D. Relies on quantitative utility.

Answer: B

Explanation

Deontology (Kant) judges by adherence to moral duties and rules. Options A and D are consequentialist. Option C is virtue ethics.

Question

Which of the following is the defining feature of virtue ethics?

- A. Rigid adherence to rules.
- B. Maximisation of utility.
- C. Focus on character traits and living a good life, with practical wisdom guiding action.
- D. Emphasis on legal compliance.

Answer: C

Explanation

Virtue ethics asks “what kind of person should I be?” and emphasises character traits (courage, justice, temperance). Options A, B, and D are distinct frameworks.

Question

Which statement is correct about the principle of justice in AI ethics?

- A. Different concepts of distributive justice (egalitarianism, utilitarianism, meritocracy) can conflict with each other.
- B. Impartial application of the law is not required for procedural justice.
- C. Personal autonomy is always secondary to social justice.
- D. Meritocracy is not a recognised distributive justice principle.

Answer: A

Explanation

Distributive justice concepts can conflict; e.g., egalitarianism vs meritocracy. Options B, C, and D are incorrect.

Question

Which statement correctly distinguishes nonmaleficence from beneficence?

- A. Nonmaleficence requires promoting others' welfare; beneficence forbids causing harm.
- B. Nonmaleficence forbids causing harm; beneficence requires actively promoting the welfare of others.
- C. They are identical.
- D. Both refer to autonomous decision-making.

Answer: B

Explanation

Nonmaleficence = do no harm; beneficence = actively help. Option A reverses. Option C conflates. Option D describes autonomy.

Question

Which of the following is the most effective way to minimise privacy risk in AI systems?

- A. Collect all possible data to improve model accuracy.
- B. Apply data minimisation and give data subjects control over their data (access, transfer, deletion rights).
- C. Store data unencrypted for analytical flexibility.
- D. Share data with third parties without consent.

Answer: B

Explanation

Data minimisation (collect only what is needed) and subject control (access, portability, deletion) are foundational privacy principles. Options A, C, and D violate them.

Question

Under GDPR, the right to be forgotten allows data subjects to:

- A. Access their personal data.
- B. Request erasure of their personal data under certain conditions.
- C. Sell their data.
- D. Demand compensation for any processing.

Answer: B

Explanation

Right to erasure (Art. 17) allows subjects to request deletion under conditions (e.g., no longer necessary, withdrawn consent). Option A describes the right of access. Options C and D misstate the right.

Question

A patient provides health data to a physician for treatment. The physician's hospital then sells the data to a marketing firm. Which principle does this violate?

- A. Data minimisation.
- B. Contextual integrity — data provided in one context (medical treatment) should not be used in an incompatible context (marketing).
- C. Right to be forgotten.
- D. Autonomy.

Answer: B**Explanation**

Contextual integrity (Nissenbaum) says data use must respect the norms of the context in which it was provided. Medical data given for treatment cannot be repurposed for marketing. Options A, C, and D describe distinct principles.

Question

Which of the following AI practices is explicitly prohibited by the EU AI Act?

- A. AI for fraud detection in banking.
- B. Social scoring by governments and manipulative AI exploiting vulnerabilities.
- C. Medical AI for diagnostic support.
- D. Chatbots for customer service.

Answer: B

Explanation

The EU AI Act prohibits social scoring, manipulative AI, exploitation of vulnerabilities, and certain biometric categorisation. Options A, C, and D are regulated but not prohibited.

Question

A US-based company provides online services to EU residents but has no physical presence in the EU. Does GDPR apply?

- A. No, GDPR only applies to EU-based organisations.
- B. Yes, GDPR has extraterritorial scope: it applies to organisations offering goods or services to EU residents or monitoring their behaviour, regardless of location.
- C. Only if the EU data subject files a complaint.
- D. Only if the company uses EU-based servers.

Answer: B

Explanation

GDPR (Art. 3) applies extraterritorially to organisations targeting EU data subjects. Options A, C, and D misstate the law.

Question

The NIST AI Risk Management Framework is organised around four functions. Which of the following correctly lists them?

- A. Train, Test, Deploy, Monitor.
- B. Govern, Map, Measure, Manage.
- C. Design, Build, Audit, Retire.
- D. Collect, Clean, Model, Predict.

Answer: B

Explanation

NIST AI RMF = Govern (structures), Map (context), Measure (assessment), Manage (mitigation). Options A, C, and D are not the framework's functions.

Question

Who are the classical proponents of utilitarianism?

- A. Kant and Aristotle.
- B. Bentham and Mill.
- C. Rawls and Nozick.
- D. Plato and Socrates.

Answer: B

Explanation

Jeremy Bentham and John Stuart Mill are the classical utilitarian philosophers. Options A, C, and D are associated with other traditions.

Question

What does Kant's categorical imperative require?

- A. Maximise aggregate utility.
- B. Act only according to maxims that could be willed as universal laws.
- C. Cultivate virtuous character.
- D. Obey the sovereign's commands.

Answer: B

Explanation

Kant's categorical imperative demands universalizability: act only on maxims you would will as universal laws. Options A and C describe utilitarianism and virtue ethics.

Question

Which two AI ethics principles are directly inherited from medical ethics?

- A. Marketing and profit.
- B. Nonmaleficence and beneficence.
- C. Privacy and censorship.
- D. Efficiency and speed.

Answer: B

Explanation

Nonmaleficence (do no harm) and beneficence (do good) are foundational medical ethics principles that have been adopted into AI ethics. Options A, C, and D are not core medical ethics principles.

Question

Why is violating privacy considered morally wrong even when no tangible harm results?

- A. It costs money.
- B. It violates individuals' rights and treats them instrumentally, as means to others' ends.
- C. It slows model training.
- D. It confuses regulators.

Answer: B

Explanation

Privacy violations wrong individuals by disrespecting their rights and autonomy, regardless of downstream harm. Options A, C, and D are beside the point.

Question

Which is a primary advantage of using synthetic data in AI development?

- A. It is always more accurate than real data.
- B. It reduces privacy risk because it does not come from real data subjects, and can mitigate bias by design.
- C. It is always cheaper to generate.
- D. It eliminates the need for validation.

Answer: B

Explanation

Synthetic data reduces privacy risk and can be designed to be balanced. Option A is false. Option C is not always true. Option D is wrong; synthetic-data models still require validation.

Question

Which is the most effective way to detect algorithmic bias?

- A. Ignore it.
- B. Regular, independent algorithmic audits combining statistical analysis with domain expertise.
- C. Use only open-source models.
- D. Remove all sensitive features.

Answer: B

Explanation

Independent algorithmic audits are the gold standard for detecting bias. Options A, C, and D are insufficient or incorrect.

Question

Why can fairness not be fully automated?

- A. Because it is too expensive.
- B. Because when base rates differ, no algorithm can simultaneously satisfy multiple fairness criteria; choosing among them is a normative judgment.
- C. Because there is not enough data.
- D. Because GDPR forbids it.

Answer: B

Explanation

Fairness involves a choice among conflicting criteria (impossibility result) and normative trade-offs that require human judgment. Options A, C, and D are not the fundamental reason.

Question

How does local explainability differ from global explainability?

- A. Local explains a specific prediction; global explains overall model behaviour.
- B. Local explains the entire model; global explains a specific case.
- C. They are identical.
- D. Local applies to linear models; global applies to nonlinear ones.

Answer: A

Explanation

Local = specific prediction (SHAP, LIME). Global = overall behaviour (feature importance, partial dependence). Options B, C, and D misstate the distinction.

Question

An applicant denied a loan is told: "If you had \$10,000 more in savings, your application would have been approved." What type of explanation is this?

- A. SHAP value.
- B. Counterfactual explanation.
- C. LIME surrogate.
- D. Global feature importance.

Answer: B

Explanation

Counterfactuals describe minimal changes to inputs that would change the output. This is actionable and intuitive. Options A, C, and D describe other techniques.

Question

Which statement best captures the accuracy-interpretability trade-off in AI?

- A. Simple models always win.
- B. Complex models (neural networks, ensembles) often achieve higher accuracy but are typically less interpretable; the trade-off should be evaluated in context.
- C. Interpretability is irrelevant in regulated industries.
- D. Complex models are always preferable.

Answer: B

Explanation

Higher flexibility often means higher accuracy but lower interpretability. The right balance depends on the use case. Options A, C, and D over-generalise.

Question

What is the maximum fine under the EU AI Act for the most serious violations?

- A. ≤1 million.
- B. ≤10 million.
- C. Up to ≤35 million or 7% of global annual turnover, whichever is higher.
- D. No fines; only warnings.

Answer: C

Explanation

The AI Act provides tiered fines, up to €35M or 7% of global turnover for prohibited practices. Options A, B, and D understate the penalties.

Question

The California Consumer Privacy Act (CCPA) gives California consumers which of the following rights?

- A. Right to know what data is collected.
- B. Right to opt out of the sale of personal information.
- C. Right to delete personal information.
- D. All of the above.

Answer: D

Explanation

CCPA grants rights to know, delete, opt out, and non-discrimination. Options A, B, and C are each part of the full set.

Question

Under GDPR, organisations must notify the supervisory authority of a personal data breach within:

- A. 24 hours.
- B. 48 hours.
- C. 72 hours.
- D. 7 days.

Answer: C

Explanation

GDPR Art. 33 requires notification within 72 hours of becoming aware of a breach. Options A, B, and D are incorrect.

Question

Which of the following obligations does the EU Digital Services Act impose on large online platforms?

- A. Preventing dissemination of illegal content.
- B. Transparency about content moderation practices.
- C. Addressing disinformation.
- D. All of the above.

Answer: D

Explanation

The DSA imposes obligations on illegal content, transparency, disinformation, algorithmic bias mitigation, and user opt-out from personalised recommendations.

Question

A key criticism of utilitarianism is that it:

- A. Ignores consequences.
- B. Can justify violating individual rights for the greater good.
- C. Emphasises character traits.
- D. Relies on rigid rules.

Answer: B

Explanation

Utilitarianism maximises aggregate welfare, which can justify rights violations. Options A, C, and D describe other frameworks or misstate the critique.

Question

Which is a standard criticism of virtue ethics?

- A. It provides no clear guidance for resolving moral dilemmas.
- B. It ignores character.
- C. It is rule-based.
- D. It only considers consequences.

Answer: A

Explanation

Critics argue virtue ethics lacks clear decision procedures and may not resolve conflicting virtues. Options B, C, and D misstate the framework.